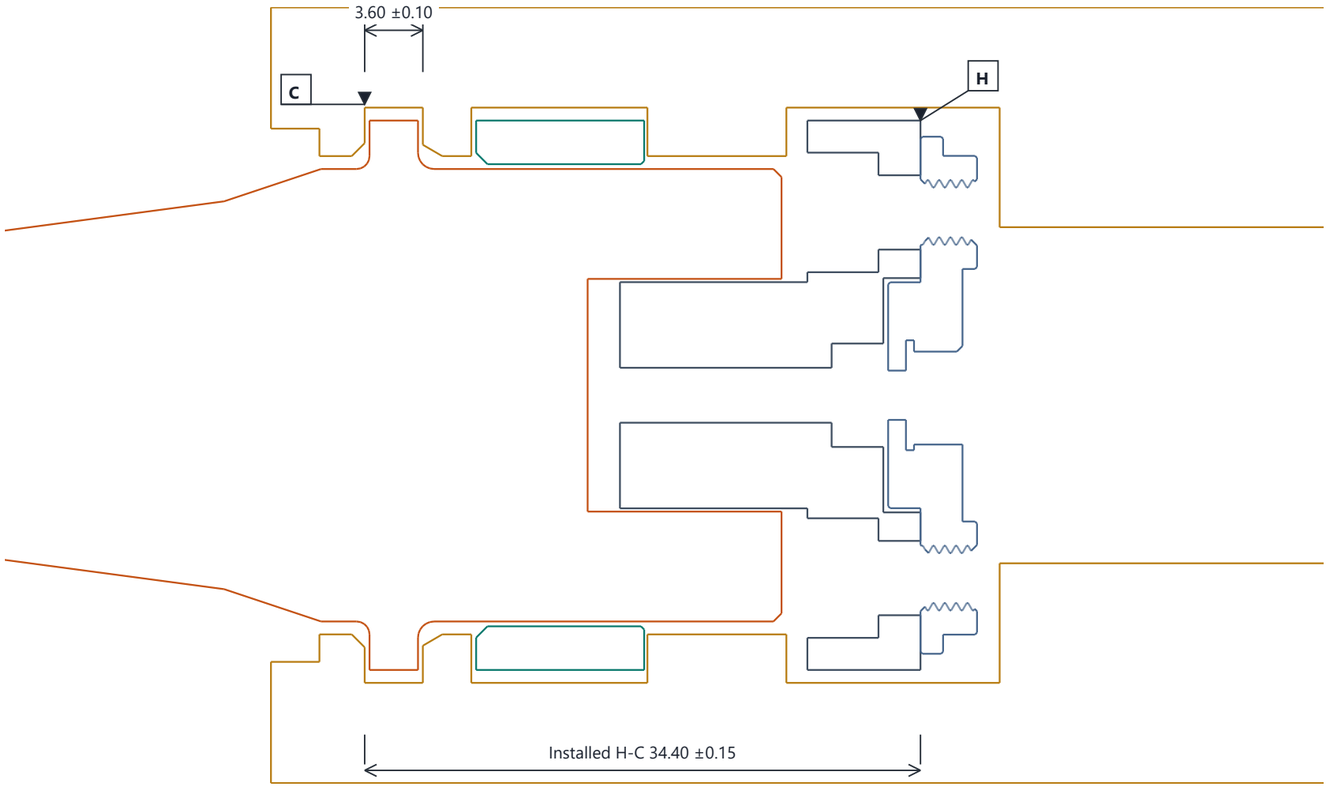
GARDEN AUGER / RELEASE B

Assembly section • retention surfaces and worst-case tolerance stack



Section through Y axis, X=0. Orange: auger. Teal: loose sleeve. Gray: adapter + metal horn.

Both chassis thrust stops act on the Ø34 integral collar. No sleeve lock, clip or separate retainer.

Assembly inspection: assembled journal axis offset ≤0.50 radial from installed servo/horn axis.
Shim/finish the printed seat if needed to achieve H-C and axis limits. Axial faces are checked at their functional extrema.

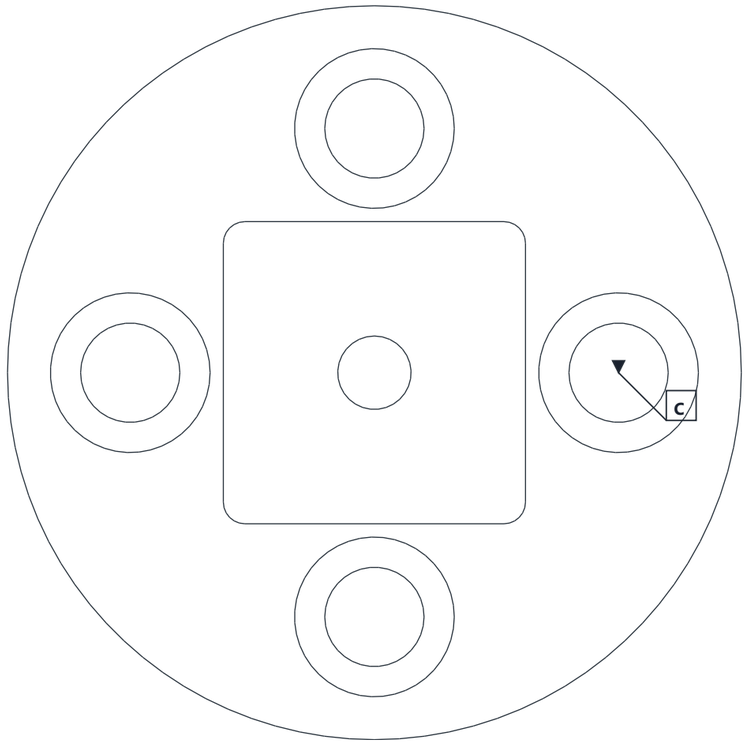
WORST-CASE ACCEPTANCE STACK

Axial end float	**0.40–0.80 mm**
Square engagement	**9.35–10.85 mm**
Tip-to-socket-bottom gap	**≥1.05 mm**
Flange-to-auger gap	**≥0.75 mm**
Journal radial clearance	**≥0.70 mm**
Allowed drive-axis offset	**≤0.50 mm radial**
Remaining radial margin	**≥0.20 mm**
Square flat clearance	**≥0.10 mm / side**
Pilot bottom clearance	**≥0.15 mm**

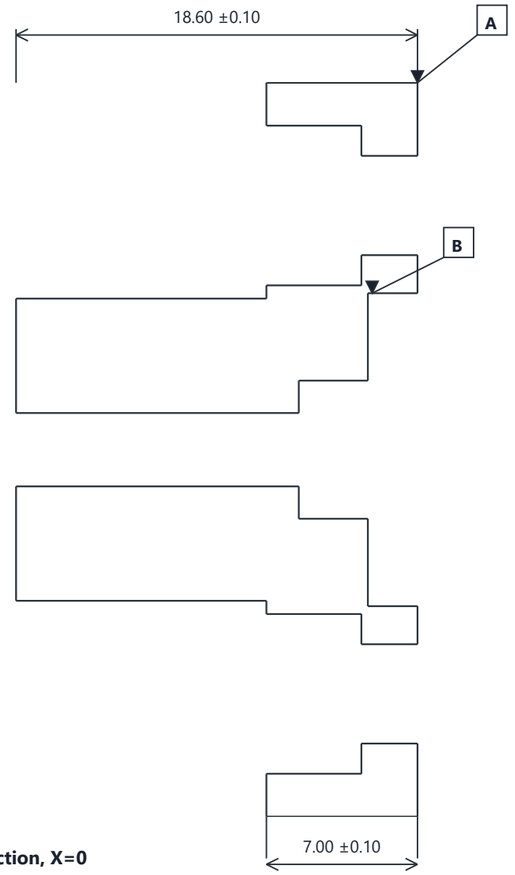
Printed critical extrema: ±0.10 mm.
Installed horn face H: ±0.15 mm from C.
Arithmetic worst case; no statistical averaging.
Limits include form within the functional envelope.
The numbers are required acceptance limits,
not measured FDM or supplier capabilities.
Qualify the fit coupon before printing the auger.

GARDEN AUGER / RELEASE B

PETG square adapter • gray #BEC3CA • dimensioned metal-horn interface



Auger-facing end view



Longitudinal section, X=0

Ø34.00 ±0.10 flange. 14.00 BASIC square, R1.00 BASIC corners.

4× Ø4.60 ±0.10 through; Ø7.40 ±0.10 counterbore, 4.40 ±0.10 deep.

Hole centers: 16 BASIC square rotated 45°; cardinal ±11.3137 BASIC.

Horn-side pilot Ø14.50 ±0.10 ×2.30 ±0.10 deep; center Ø3.40 through.

Center head relief Ø6.40, front at Y142.60; seats the horn M3 head clear.

M3×6: seat-to-shaft-tip 2.60 ±0.20; require ≥4.00 usable thread depth.

⌓	0.10	**A**	**B**	**C**	Square surfaces, all around
⌖	Ø0.20	**A**	**B**		4× mount holes, RFS
▱	0.10				Face A

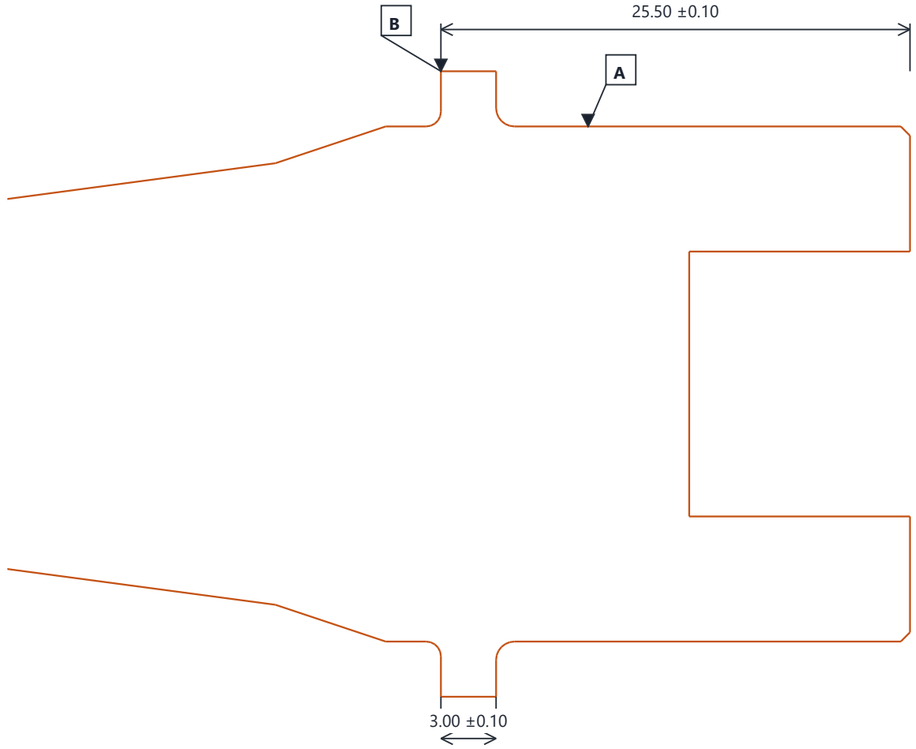
A: horn seating plane. B: pilot bore axis.

C: +X mounting hole, used only to clock the square.

Basic square sizes use profile; no extra size tolerance added.

GARDEN AUGER / RELEASE B

PETG auger journal • orange #D57E37 • loose PLA sleeve • teal #37AAA1



Tail section; continuous Ø28 journal axis is datum A





Full slip-on sleeve; no key or lock

Journal Ø28.00 ±0.10. Collar Ø34.00 ±0.10 ×3.00 ±0.10.

Socket 14.40 BASIC across flats, R1.20 BASIC; depth12.00 ±0.10.

Socket is concentric to datum A; clocking is free before assembly.

Keep the modeled collar fillets and 0.50 outer mouth chamfer.

Nominal minimum hub corner wall: 4.31 straight / 3.81 at mouth.

Geometric backlash about 3.89° total; this is not a zero-play joint.

⌓	0.10	**A**	**B**	Socket surfaces, all around
⊥	0.10	**A**		Collar thrust faces; within axial size envelope

Sleeve OD Ø34.00 ±0.10; bore Ø28.60 ±0.10.

Length10.40 ±0.10; lead-ins match CAD.

Housing sleeve cavity Ø35.60 ±0.10.

Combined minimum radial float0.90 mm.

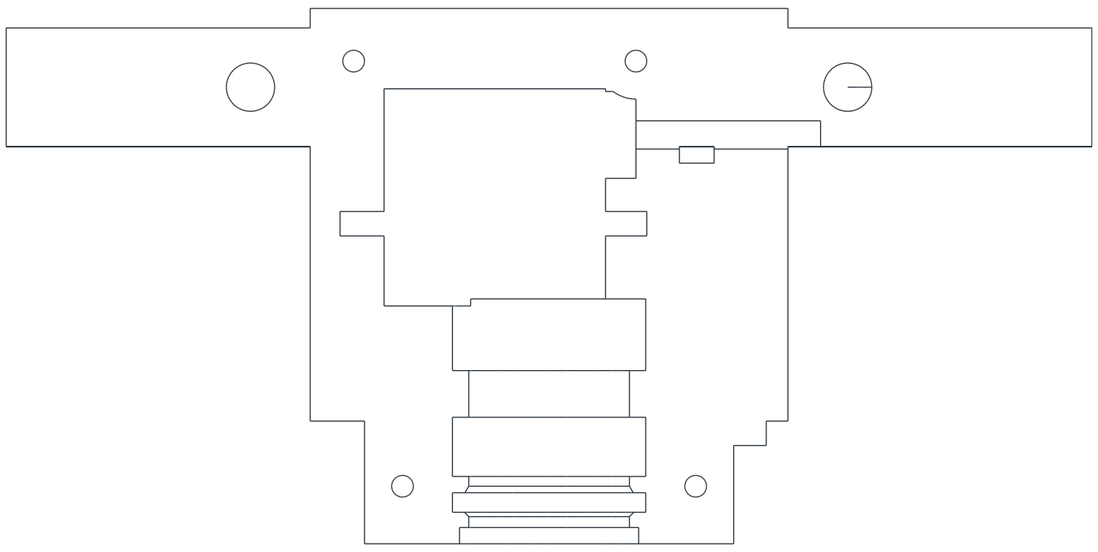
Sleeve slides over the journal from the drive end.

PLA/PETG wear benefit needs a physical trial.

One-piece auger export includes the original flight.

GARDEN AUGER / RELEASE B

PETG chassis • yellow #E5AE3C • datums, insert pockets and fasteners



Lower housing, split-face view. Upper uses matching screw pattern.

HOUSING DATUM REFERENCE FRAME

A: machined/finished clamshell split plane (Z=0).

B: common assembled Ø29.60 journal axis.

C: front collar thrust-stop plane (Y113.70).

Thrust-stop span3.60 ±0.10. Cavity Ø35.60 ±0.10.

Journal Ø29.60 ±0.10; front/rear axes must be common.

Inspect with both chassis halves clamped together.

▱	0.10		Split plane A
⊥	0.10	**B**	Thrust stops, within axial span limits

FOUR HEAT-SET POCKETS

ruthex RX-M3x5.7 brass: Ø4.6 ×5.7, M3 thread.

Pilot Ø4.00 +0.10/-0.00 ×6.70 +0.20/-0.00 deep.

Install flush; pocket depth ≥actual insert length +1.00.

Front centers: X±27.00, Y118.50.

Rear centers: X−36.00 / +16.00, Y196.80.

Upper Ø3.40 clearance / Ø6.20 head recess.

2× M3×12 front; 2× M3×18 rear, ISO4762.

Thread engagement4.00 front /5.30 rear nominal.

Rear fastener pads clear the relocated wire channel.

Checked insert-wall margin: minimum 4.45 mm after dimensional allowance (required ≥1.60 mm).

Material labels and colors are set in native CAD and 3MF. Choose actual printer, filament and support profiles in the slicer.